

# Probability Models, Chapter 1:

## Solutions to Various Exercises

Nolan R. H. Gagnon

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**1.** A box contains three marbles: one red, one green, and one blue. Consider an experiment that consists of taking one marble from the box then replacing it in the box and drawing a second marble from the box. What is the sample space? If, at all times, each marble in the box is equally likely to be selected, what is the probability of each point in the sample space?

**Solution:** Let  $(x, y)$  denote the outcome of choosing a marble of color  $x$  on the first draw and a marble of color  $y$  on the second draw, where  $x, y \in \{\text{red, green, blue}\}$ . The sample space is

$$\begin{aligned}\Omega = \{ & (\text{red, red}), (\text{red, green}), (\text{red, blue}), \\ & (\text{green, red}), (\text{green, green}), (\text{green, blue}), \\ & (\text{blue, red}), (\text{blue, green}), (\text{blue, blue}) \}.\end{aligned}$$

Since the two draws are independent, the probability of each point in the sample space is  $\frac{1}{9}$ .

**2.** Repeat Exercise 1 when the second marble is drawn without replacing the first marble.

**Solution:** In this case, the sample space is

$$\begin{aligned}\Omega = \{ & (\text{red, green}), (\text{red, blue}), \\ & (\text{green, red}), (\text{green, blue}), \\ & (\text{blue, red}), (\text{blue, green}) \}.\end{aligned}$$

The probability of any point  $(x, y)$  in the sample space is  $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$ .

**3.** A coin is to be tossed until a head appears twice in a row. What is the sample space for this experiment? If the coin is fair, what is the probability that it will be tossed exactly four times?

**Solution:** Let  $H$  denote getting heads on a coin flip, and let  $T$  denote getting tails. Then the sample space is

$$\Omega = \{HH, THH, TTHH, \dots, HTHH, HTTHH, \dots\}.$$

The probability that the coin will be tossed four times is

$$P(TTHH) + P(HTHH) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

4. Let  $E, F, G$  be three events. Find expressions for the events that of  $E, F, G$

- (a) only  $F$  occurs,
- (b) both  $E$  and  $F$  but not  $G$  occur,
- (c) at least one event occurs,
- (d) at least two events occur,
- (e) all three events occur,
- (f) none occurs,
- (g) at most one occurs,
- (h) at most two occur.

**Solution:** The expressions are

- (a)  $FE^cG^c$
- (b)  $EE^cFG^c$
- (c)  $E \cup F \cup G$
- (d)  $EF \cup EG \cup FG$
- (e)  $EEFG$
- (f)  $E^cF^cG^c$
- (g)  $EF^cG^c \cup E^cFG^c \cup E^cF^cG$
- (h)  $EEFG^c \cup E^cEFG \cup EE^cG$ .

4. An individual uses the following gambling system at Las Vegas. He bets \$1 that the roulette wheel will come up red. If he wins, he quits. If he loses then he makes the same bet a second time, only this time he bets \$2; and then regardless of the outcome, quits. Assuming that he has a probability of  $\frac{1}{2}$  of winning each bet, what is the probability that he goes home a winner? Why is this system not used by everyone?

**Solution:** Let  $W_1$  denote the winning the first bet, and let  $W_2$  denote winning the second bet. The probability that he goes home a winner is

$$\begin{aligned} P(\text{WINNER}) &= P(W_1) + P(W_2|W_1^c) \\ &= \frac{1}{2} + \frac{P(W_2|W_1^c)}{P(W_1^c)} \\ &= \frac{1}{2} + \frac{1/2}{1/2} \\ &= \frac{3}{4}. \end{aligned}$$

++++++The reason this strategy isn't used by everyone is because it has an expected net earning of \$0. To see this, let  $X$  be a random variable representing the amount of money the gambler nets using this strategy. There are three possible values that  $X$  can take. If he wins the first round, he nets \$1. If he loses the first round, but wins the second round, he nets  $-1 + 2 = 1$  dollar. If he loses the first round and loses the second round, he nets  $-1 - 2 = -3$  dollars. The support of  $X$  is thus  $\{1, -3\}$ . As shown previously, the probability of winning 1 dollar is  $\frac{3}{4}$ . The probability of losing 3 dollars must then be  $\frac{1}{4}$ . Therefore, the expected net earning is

$$E[X] = 1 \times \frac{3}{4} - 3 \times \frac{1}{4} = 0.$$

**10.** Prove that Boole's Inequality holds, i.e., that

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i).$$

**Proof:** This proof is broken into two parts.

**Part 1:** It can be shown that  $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i$ , where  $F_1 = E_1$  and  $F_i = E_i \cap \bigcap_{j=1}^{i-1} \overline{E_j}$  for  $i \geq 2$ . To this end, let  $x \in \bigcup_{i=1}^n E_i$ . Then  $x$  is an element of at least one of the  $E_i$ . For the sake of reaching a contradiction, suppose that  $x \notin \bigcup_{i=1}^n F_i$ . The implication of this supposition is that  $x \in \bigcap_{i=1}^n (\overline{E_i} \cup \bigcup_{j=1}^{i-1} E_j)$ . Thus,  $x \in \overline{E_1}$ , and  $x \in \overline{E_2} \cup E_1$ , and  $x \in \overline{E_3} \cup E_2 \cup E_1$ , and so on. But considering each of these unions in order leads to the conclusion that  $x \notin E_1$  and  $x \notin E_2$  and  $x \notin E_3$ , and so on. Therefore,  $x$  cannot be an element of the set  $\bigcup_{i=1}^n E_i$ . This is a contradiction, so  $x \in \bigcup_{i=1}^n F_i$ .

The reverse set inclusion is trivially proven. Thus,  $\bigcup_{i=1}^n F_i = \bigcup_{i=1}^n E_i$ .

**Part 2:** Using the result of part one of this proof,  $P(\bigcup_{i=1}^n E_i)$  can be written as

$$P\left(\bigcup_{i=1}^n E_i\right) = P\left(\bigcup_{i=1}^n F_i\right) = P\left(\bigcup_{i=1}^n \left[E_i \cap \bigcap_{j=1}^{i-1} \overline{E_j}\right]\right).$$

Clearly, the  $F_i$  are mutually exclusive. Therefore, the  $\sigma$ -additivity axiom of probability distributions applies, i.e.,

$$P\left(\bigcup_{i=1}^n \left[E_i \bigcap_{j=1}^{i-1} \overline{E_j}\right]\right) = \sum_{i=1}^n P\left(E_i \bigcap_{j=1}^{i-1} \overline{E_j}\right) \quad (1)$$

$$= P(E_1) + P(E_2 \overline{E_1}) + \dots + P(E_n \overline{E_{n-1}} \dots \overline{E_1}) \quad (2)$$

$$= \sum_{i=1}^n P(E_i) + \sum_{j=1}^{n-1} \left[ P\left\{\overline{\bigcup_{k=1}^j E_k}\right\} - P\left(E_{j+1} \bigcup \left\{\overline{\bigcup_{k=1}^j E_k}\right\}\right)\right]. \quad (3)$$

Now, clearly the set  $E_{j+1} \bigcup \left\{\overline{\bigcup_{k=1}^j E_k}\right\}$  is larger than or equal to the set  $\overline{\bigcup_{k=1}^j E_k}$ . Hence, the second summation in equation (3) is less than or equal to 0. It can, therefore, be concluded that  $P(\bigcup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i)$ . This completes the proof. ■